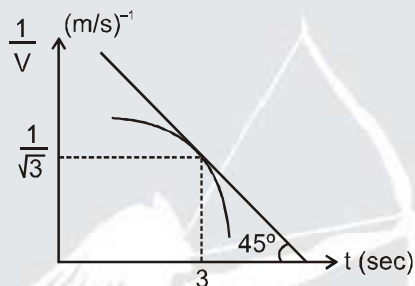
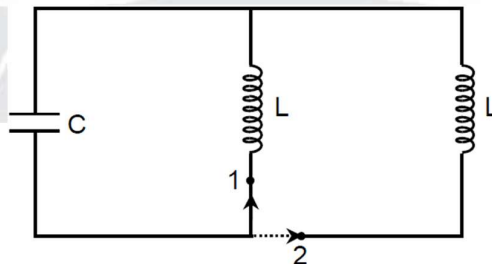


1. The graph shows the variation of $\frac{1}{V}$ (where V is the velocity of the particle) with respect to time. Then find the value of acceleration at $t = 3$ sec.

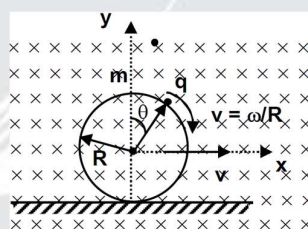


- (A) 3 m/s^2 (B) 5 m/s^2 (C) 1 m/s^2 (D) $3\sqrt{3} \text{ m/s}^2$
2. There exist a uniform magnetic field $\vec{B} = 10T \hat{k}$ in space. A point charge (specific charge $\frac{q}{m} = 1C/\text{kg}$ is projected from $(-5 \text{ m}, 0, 0)$ with velocity $\sqrt{3}v \hat{i} + v \hat{j}$. The value of 'v' for which the charge particle will pass through origin.
- (A) 5m/s (B) 10m/s (C) 15m/s (D) 25m/s
3. A block is kept on a smooth inclined plane of angle of inclination θ that moves with a constant horizontal acceleration so that the block does not slide relative to the inclined plane. If the inclined plane stops, the normal contact force offered by the plane on the block changes by the factor
- (A) $\cos\theta$ (B) $\sec^2\theta$ (C) $\cos^2\theta$ (D) $\sin^2\theta$
4. A capacitor with charge Q on it is connected to an inductor L as shown in diagram at $t = 0$. When the switch is flipped from position 1 to 2, the current in the circuit is observed to be at half of its maximum value. Then

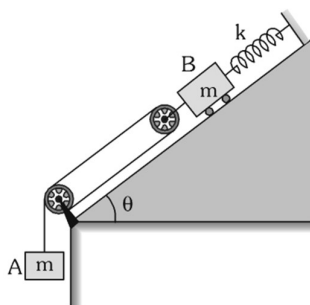


- (A) Charge on capacitor at the time was $Q/2$ (B) Charge on capacitor at that time was $\frac{Q\sqrt{3}}{2}$
- (C) Total energy lost is $\frac{Q^2}{4C}$ (D) Total energy lost is $\frac{Q^2}{8C}$

5. Two objects of mass m and $4m$ are at rest at an infinite separation. They move towards each other under mutual attraction. If G is the universal gravitational constant. Then at separation r .
- (A) The sum of energy of the two objects is negative.
- (B) Their relative velocity of approach is $\left(\frac{10Gm}{r}\right)^{1/2}$ in magnitude
- (C) The sum of kinetic energy of the objects is $\frac{4Gm^2}{r}$.
- (D) The sum of angular momentum of both the objects is zero about any point.
6. A ring of mass m and radius R is set into pure rolling on horizontal rough surface, in a uniform magnetic field of strength B as shown in the figure. A point charge of negligible mass is attached to rolling ring. Friction is sufficient so that it does not slip at any point of its motion. (θ is measured in clockwise from positive y-axis)



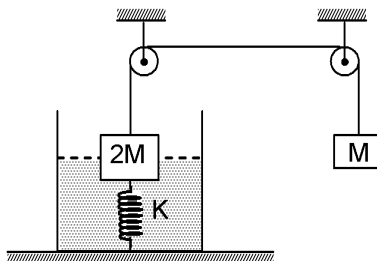
- (A) Ring will continue to move with constant velocity.
- (B) The value of friction acting on ring is $Bqv \cos \theta$.
- (C) The value of friction acting on ring is $Bqv \sin \theta$.
- (D) Ring will lose contact with ground if v is greater than $\left(\frac{mg}{2qB}\right)$
7. Pulleys are massless and friction is absent everywhere.



Choose the correct statement(s) :

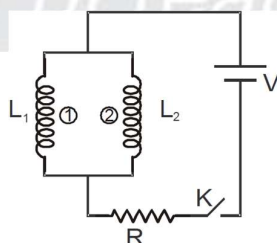
- (A) The natural angular frequency of system is $\sqrt{\frac{k}{5m}}$
- (B) The natural angular frequency of system is $\sqrt{\frac{k}{7m}}$
- (C) The natural angular frequency of system is $\sqrt{\frac{k}{3m}}$
- (D) The two blocks oscillate at same frequency.

8. A uniform cylindrical block of mass $2M$ and cross-sectional area A remains submerged in a liquid of density ρ , density of the material of the cylinder is 3ρ . The cylinder is connected to lower end of the tank by means of a light spring of spring constant K . The other end of the cylinder is connected to another block of mass M by means of a light inextensible string as shown in the figure. The pulleys shown are massless and frictionless and assume that the cross-section of the cylinder is very small when compared to that of the tank. Under equilibrium conditions, half of the cylinder is submerged.



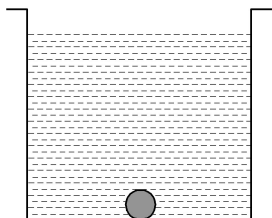
Choose the correct statements

- (A) Under equilibrium conditions, compression of the spring is $\frac{2Mg}{3K}$.
- (B) The cylinder is pushed down and then released. The maximum distance the cylinder can be lowered before being released, if the string were not become slack in the subsequent motion, is $\frac{3Mg}{(K + A\rho g)}$
- (C) The cylinder is pushed down and then released. The maximum distance the cylinder can be lowered before being released, if the string were not become slack in the subsequent motion, is $\frac{2Mg}{(K + A\rho g)}$
- (D) If the cylinder is pushed down from equilibrium by a very small distance (the string remains taut), the time period of subsequent motion is $2\pi \frac{3M}{K + A\rho g}$
9. In the figure shown the key is switched on at $t = 0$. Let I_1 and I_2 be the currents through inductors having self inductances L_1 & L_2 at any time t respectively. The magnetic energy stored in the inductors 1 and 2 be U_1 and U_2 . Then $\frac{U_1}{U_2}$ at any instant of time is :



- (A) $\frac{L_1}{L_2}$ (B) $\frac{L_2}{L_1}$ (C) $\frac{I_1}{I_2}$ (D) $\frac{I_2}{I_1}$

10. Two spherical planets P and Q have the same uniform density ρ , masses M_P and M_Q , with surface areas A and $4A$, respectively. A spherical planet R also has uniform density ρ and its mass is $(M_P + M_Q)$. The escape velocities from the planets P, Q and R, are V_P , V_Q and V_R respectively. Then
- (A) $V_Q > V_R > V_P$ (B) $V_R > V_Q > V_P$ (C) $V_R/V_P = 3$ (D) $V_P/V_Q = 1/2$
11. A small metal piece of density $\frac{2\rho_0}{3}$ is held at the bottom of a container. Containing liquid whose density varies with height h from the bottom as $\rho = \rho_0 - \beta h$, where β is a positive constant. At $t = 0$, the piece is released from rest. Assume that the liquid is present in the container upto a level such that piece always remains inside the liquid.



Also given $\rho_0 = 5000$ SI unit

$\beta_0 = 3000$ SI unit

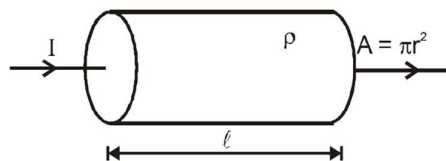
$g = 10 \text{ m/s}^2$

Choose the correct statement(s) :

- (A) The maximum height reached by piece in liquid is $\frac{10}{9} \text{ m}$
- (B) The maximum velocity of the piece inside liquid is $\left(\frac{5}{3} \text{ m/s}\right)$
- (C) The maximum acceleration of piece inside liquid is 5 m/s^2
- (D) The maximum acceleration of piece inside liquid is 8 m/s^2

Comprehension

Suppose $\vec{E}$ and $\vec{B}$ are electric field and magnetic field at a point in space. The vector $\frac{\vec{E} \times \vec{B}}{\mu_0}$ is known as the Poynting vector. The magnitude of Poynting vector is the rate of flow of energy per unit area per unit time. The rate at which electromagnetic energy flows through any area is the flux of Poynting vector through that area. Consider a cylindrical conductor of length l and radius of cross-section r . It carries I current distributed uniformly through out its cross section. Resistivity of conductor is ρ .



12. What is magnitude of Poynting vector at a point on lateral surface of the conductor ?
- (A) $\frac{I\rho}{2\pi r^2}$ (B) $\frac{I^2\rho}{2\pi r^3}$ (C) $\frac{I^2\rho}{2\pi^2 r^2}$ (D) $\frac{I^2\rho}{2\pi^2 r^3}$

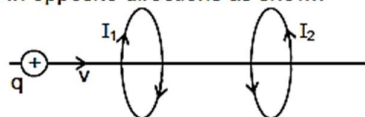
13. Total energy radiated per unit time through half cylindrical surface of the conductor is –

- (A) $\frac{I^2 \rho \ell}{\pi r^2}$ (B) $\frac{I^2 \rho \ell}{\pi^2 r^3}$ (C) $\frac{I^2 \rho \ell}{2\pi r^2}$ (D) Zero

14. The figures in list-I show some charge and current distribution with a charged particle projected in some specific direction list-II gives certain conditions which may exist in the subsequent motion of the charged particle. Match list I with list II and select the correct answer using the codes given below the lists : (Neglect gravity force on charged particle)

List-I

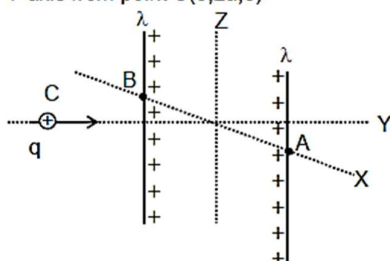
- (P) A positive charge projected along the axis of two coaxial current carrying loops, carrying currents in opposite directions as shown



List-II

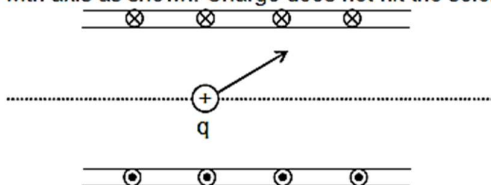
- (1) Magnitude of acceleration of the charged particle is constant

- (Q) Two long line charges having equal charge density λ parallel to z-axis passing through points A (a,0,0) and B (-a,0,0) and a positive charge projected along Y-axis from point C(0,2a,0)



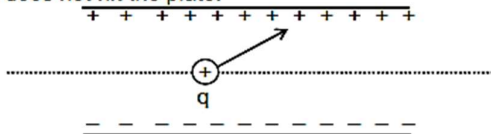
- (2) Kinetic energy of the charged particle is constant

- (R) A positive charge is projected from an inside point on the axis of a long solenoid at some angle θ with axis as shown. Charge does not hit the solenoid



- (3) Angular momentum of the charged particle about the point of projection is constant

- (S) A positive charge projected from a point in between two large parallel oppositely charged horizontal plates at some angle θ with horizontal as shown. Charge does not hit the plate.

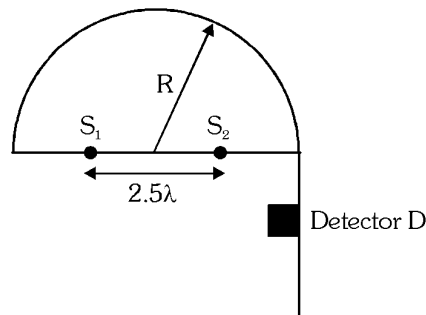


- (4) Path of the charged particle is straight

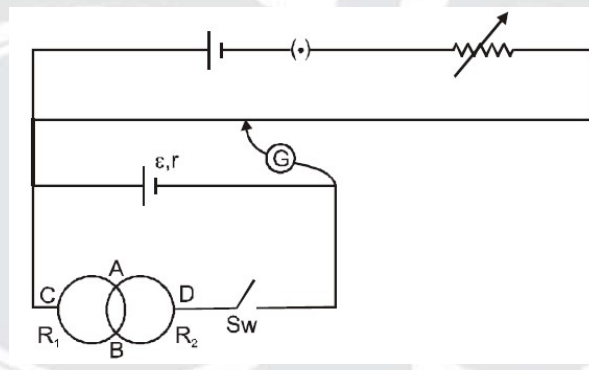
Codes :

	P	Q	R	S
(A)	3	1	2	4
(B)	3	2	1	4
(C)	4	2	1	3
(D)	3	4	2	1

15. Two in-phase, coherent point sources of sound are shown in figure. A sound detector, D is moving on a semi-infinite vertical wall connected to a semicircular portion (centred at the mid point of the line joining the two sources). Wave length of sound waves is λ and separation between the point sources is 2.5λ . The radius of the semi-circle $R > \lambda$. The number of minimas recorded by the detector D on the track are N, find N.

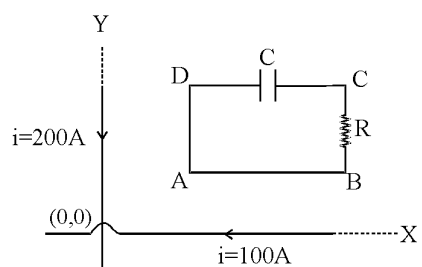


16. Internal resistance of a cell is $r = 5\Omega$ and it is connected in potentiometer (wire length = 1m) circuit arrangement as shown in figure, which shows two circular conducting rings R_1 & R_2 each having radii 20 cm crossing each other at conducting joints A & B. Section AB subtends 120° at the center of each ring. Resistance per unit length of R_1 & R_2 are $\frac{3}{\pi} \Omega/\text{cm}$ & $\frac{6}{\pi} \Omega/\text{cm}$ respectively. Cell is connected across points C & D of rings lying on perpendicular bisector of AB. Now when switch Sw remains closed, balancing length at null point is 48 cm. Find the balancing length (in cm.) when switch Sw is opened.



17. A gas containing hydrogen like ions with atomic number Z , emits photons in transition $(n + 2) \rightarrow n$, where $n = Z$. These photons fall on a metallic plate and eject electrons having minimum de-Broglie wavelength λ of $5\text{\AA}$. Find Z . (the work function of metal is 4.2 eV)
18. Solid uniform conducting sphere of mass ' m ' and charge Q , rotates about its axis of symmetry with constant angular velocity ' ω ' then the ratio of magnetic moment to the moment of inertia of the sphere is $\frac{xQ\omega}{6m}$ then x is : (Neglect induced charges due to centrifugal force)

19. Using dimensional analysis, angular frequency (ω) of surface waves on a lake as a function of surface tension σ , water density ρ and the wave number $k = \frac{2\pi}{\lambda}$ (λ is the wavelength) is derived. Assuming that the dimensionless coefficient is equal to two (in SI units) in this case, find the value of group velocity (v_g) where $v_g = \frac{\partial \omega}{\partial k}$. If your answer is v_s (in m/s) then fill value of $10v_s$. (Use : $\sigma = 64 \text{ mN.m}^{-1}$, $\rho = 320\pi \text{ kg.m}^{-3}$, $\lambda = 1 \text{ cm}$)
20. Two straight infinitely long current wires lying along the x-axis and the y-axis carry currents 100 A and 200 A as indicated in the adjoining figure. A rigid square metallic wire frame ABCD of side 10 cm and resistance ($R = 10\Omega$) is connected to a capacitor $C_1 = 1\text{mF}$ as shown in the figure. The wire frame, which lies in the x-y plane, moves with a constant velocity $30(\hat{i} + \hat{j}) \text{ m/s}$. At the instant, when the point A is located at (20 cm, 20 cm) the charge q_0 on the capacitor (as shown in the figure) is $0.2 \mu\text{C}$. If, at the above mentioned instant, the current flowing in the Circuit is given by 10^{-n} amp . Fill the value of n in nearest integer.



ANSWER KEY

1. (A)	2. (D)	3. (D)	4. (BD)	5. (BCD)
6. (ACD)	7. (AD)	8. (ABD)	9. (BC)	10. (BD)
11. (ABC)	12. (D)	13. (D)	14. (D)	15. 8
16. $\ell = 52$	17. 2	18. 5	19. 6	20. 5